

SSD MEMORY DESIGN

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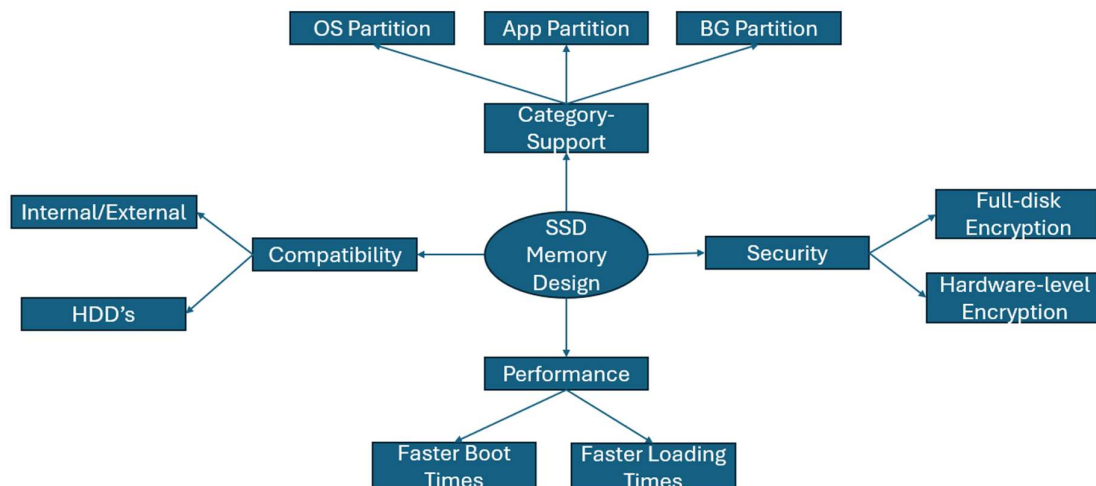
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Introduction

This SSD Memory Design is designed to improve performance on both internal and external SSDs. It is also designed to improve performance of older hardware such as internal and external HDDs. This SSD memory design was developed to allow for quicker loading times and boot times on modern computing systems that require performance and reliability. This SSD memory design does not require any direct hardware changes and can be implemented quickly through OS and minor firmware updates. This design will explain the category-based memory, the workflow and the trade-offs with migrations.

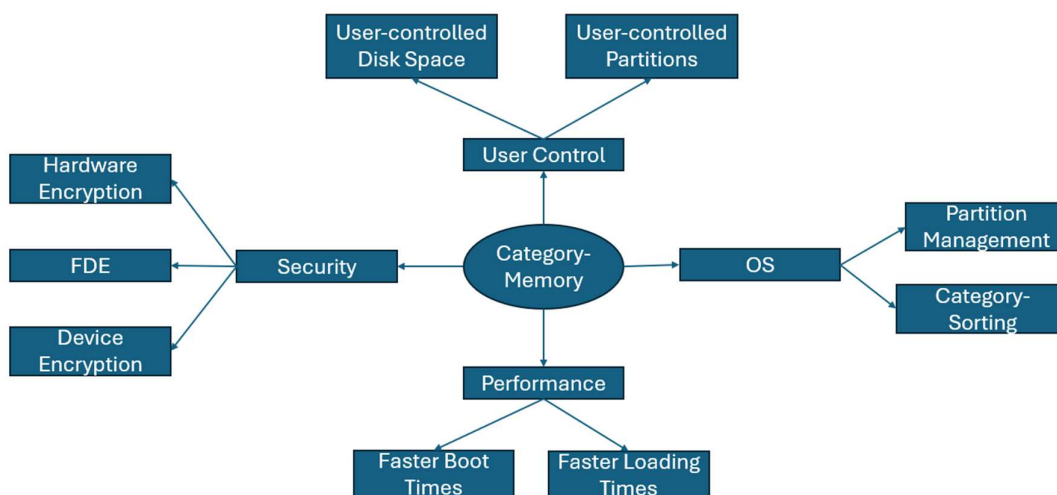
This SSD Memory Design cannot promise perfect performance or reliability. However, it can promise to improve performance through loading and boot times on modern computing systems. It can also provide compatibility with HDDs both externally and internally as this memory design is heavily managed by the operating system. It is important to remember that this memory design may be less effective on HDD's compared to SSDs. Make sure to adapt any features to the system's need.



The spider diagram above shows a visual layout of the SSD Memory Design. It shows 2-3 features of each principle involved in the design. This diagram does not show all features, and more detail can be found in the sections below.

SSD Memory Design Category-Based Memory

- Category-based memory in an SSD or HDD involves splitting the data into partitions.
- This can be done by the operating system as they already manage partitions on hard drives and would just need to be configured to sort data based on category
- These partitions could be Operating System, Apps and Background.
- This also means that internal and external hard drives can be sorted by the OS for quicker access reducing load times.
- This category-based memory can mainly use a fixed priority system for allocation space where certain categories may be given more space than others.
- For example, on a gaming PC the application partition is allocated the most space.
- Operating systems may allow for a dynamic and user-controlled priority system as users could be able to allocate disk space for partitions.
- Users should also still be allowed to create their own partitions on the disk and allocate specific disk space but with a warning that this reduces space for the main categories
- For security, this SSD memory design does not need to provide a new encryption mode due to pre-existing security systems.
- For example, operating systems and devices offer full-disk encryption, hardware-level encryption and device encryption.
- OS vendors would be able to implement this memory design in the next few update cycles as no direct hardware and minor firmware changes are needed for the DMA.
- This category-based memory could improve boot times and loading times in modern computing systems heavily



The spider diagram above shows a visual layout of the category-based memory for both SSDs and HDDs. This does not show the exact details of the category-based memory architecture. Further details can be found in the section above the diagram.

SSD Memory Design Workflow

- The workflow will show how the SSD design should work in a device and its connection to RAM, CPU, Cache and the GPU.
- For the first boot of the device, BIOS and other bootloader data are loaded from the hard drive without accessing any partitions as they don't exist yet
- The OS is then loaded from the SSD into RAM by the CPU and DMA by checking the whole disk space.
- After the OS is loaded, it sorts out all data on the device in the OS, App and Background partitions for quicker access.
- For the second boot, BIOS and other bootloader data are loaded from SSD into RAM to start up the device.
- This is done as the CPU and DMA can quickly access data from the OS partition leading to faster access times.
- OS is loaded from SSD into RAM to load the operating system into the device for the user to use.
- This is also done by the GPU and DMA as they can continue to quickly access data from the OS partition.
- This leads to faster boot times on devices especially those with NVMe SSDs for internal hard drives.
- The user starts a gaming application on the device after the second boot.
- For example, this gaming application could be GTA 6 or COD.
- The application is stored within the app partition where GTA-6 specific data is moved to RAM as instructions by the CPU and DMA
- The operating system then organises GTA-6 specific instructions into the application category to allow for the CPU to access it quicker.
- The most frequently used GTA-6 data and instructions are stored in L1 or L2 cache depending on the priority system that the device is configured with.
- The CPU can then access GTA-6 specific instructions quickly leading to faster loading times.
- GTA-6 3D Geometry, Images and Rendered Frames are moved to VRAM via the CPU's PCI bus.
- The operating system then organises the GTA-6 VRAM data into the 3 categories of Physics Model, Textures and Frame Buffers.
- L2 cache will prioritize Physics Models as it would be the heaviest workload of GTA 6.
- Each L1 cache and SMs/CUs will be dedicated to a specific category in VRAM with physics models being the main priority.
- Results are directly written to VRAM and then sent to the CPU for processing where they are then displayed on the user's monitor/screen.

SSD Memory Design Trade-Offs

- This SSD memory design has its trade-offs just like any other system I have created.
- There may also be more implementation challenges although they should be relatively lighter compared to the GPU, RAM and Cache designs.
- I have identified 3 main trade-offs for the SSD Memory Design.
- The first trade-off is heat.
- This trade-off exists since DMA and CPU can access data from SSDs more quickly which can produce more heat.
- This means that without heat management the SSD can overheat causing issue for the user.
- This trade-off can be managed by hard drive-specific cooling methods such as passive cooling and general device cooling systems.
- The second trade-off is first-boot setup.
- This trade-off exists because the organised partitions have not been created by the operating system yet.
- This means that boot times may be slower as the CPU and DMA can take longer to access data from the SSD.
- This trade-off is acceptable as it is a one-time cost and external hard drives can completely avoid this trade-off as they don't rely on boots.
- The third trade-off is performance inequality.
- This trade-off exists since SSDs especially NVMeS can perform more effectively due to faster read/write speeds compared to HDDs.
- This means that boot times and loading times may be slower on a device with an internal HDD compared to a device with an internal SSD.
- This trade-off is acceptable as HDDs are accepted by users as older hardware and it is managed as the memory architecture is also available for HDDs.
- The fourth trade-off is OS-Reliance
- This trade-off exists because the OS must perform partition management, key management, and category-sorting
- This means that the effectiveness of this SSD memory architecture may be limited by OS limitations.
- This trade-off is acceptable as operating systems act as the boundary between software and hardware, and it is also managed through other systems like DMA and the CPU.